

Proposal 1: Airline Flight Delay Risk Prediction and Operational Prioritization

The original proposal has strong potential because it combines official BTS flight data, descriptive and diagnostic analytics, machine-learning prediction, SHAP explainability, optimization, and a decision-support interface. However, its current scope is too broad for eight weeks, and several proposed predictors would cause target leakage because they are only known after departure or after the delay occurs.

The following version preserves the project's technical strength while making it methodologically valid and achievable.

1. Refined project title

AI-Powered Flight Delay Risk Prediction and Operational Prioritization Using U.S. Bureau of Transportation Statistics Data

An alternative shorter title is:

Flight Delay Risk Intelligence and Operational Prioritization

2. Refined project purpose

The purpose of the project is to develop a prototype decision-support system that:

1. Analyzes historical airline delay patterns.
2. Identifies operational and scheduling factors associated with arrival delays.
3. Predicts the probability that a scheduled flight will arrive at least 15 minutes late.
4. Explains individual and overall predictions using SHAP.
5. Prioritizes high-risk flights for operational review when intervention capacity is limited.
6. Presents results through an interactive dashboard.

The project must be framed as a **schedule-time prediction system**. This means that the prediction is generated before the flight departs using only information that would reasonably be known when the flight is scheduled or during predeparture planning.

3. Refined business problem

Airline operations managers cannot provide additional attention to every scheduled flight. They need a method for identifying flights that are most likely to experience significant arrival delays.

The project will answer the following business question:

Given a set of scheduled flights and limited operational-review capacity, which flights should be prioritized because they have the highest predicted risk of arriving at least 15 minutes late?

The proposed system will not claim to eliminate delays. Instead, it will help managers:

- Identify high-risk flights
 - Understand the factors contributing to the risk
 - Allocate limited operational attention more systematically
 - Compare risk across airlines, airports, routes, and time periods
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4. Dataset requirements

Primary dataset

Use the official **Airline On-Time Performance Data** published by the U.S. Bureau of Transportation Statistics.

The original proposal identifies this official dataset and describes variables related to airlines, airports, schedules, delays, cancellations, diversions, and delay causes.

Study period

Do not use partial 2026 data as the main modelling dataset.

Use one of the following approaches:

Preferred approach

Use the latest two complete calendar years available at the beginning of the project.

- Earlier complete year: training and validation
- Most recent complete year: final testing

Acceptable alternative

Use one complete calendar year and apply a chronological split:

- Earlier months: training
- Later month or months: validation
- Final month or months: testing

The group must not use a random split as the only validation method because flight operations contain temporal patterns.

Dataset size

The BTS dataset may contain millions of records. The group should:

- Download only required columns
- Use efficient data types
- Process monthly files separately before combining them
- Consider Polars, DuckDB, PyArrow, or chunked Pandas processing
- Document any filtering or sampling

A derived analytical subset is acceptable, but it must be clearly described as a subset of the official dataset. The group must not generate or simulate flight records.

5. Target variable

Primary prediction target

Use:

ArrDel15

where:

- 1 indicates an arrival delay of at least 15 minutes
- 0 indicates an arrival delay of less than 15 minutes

The primary machine-learning problem is therefore:

Binary classification of arrival delay risk.

Optional secondary target

Only if the primary project is completed successfully, the group may develop a regression model for:

ArrDelay

This secondary model would predict the number of arrival-delay minutes.

The regression model is optional and must not reduce the quality of the required classification model.

6. Critical target-leakage restrictions

The original proposal includes several variables that cannot be used in a true predeparture prediction model.

The following variables must be excluded from the schedule-time predictive model:

- DepDelay
- DepDel15
- Actual departure time
- Actual arrival time
- TaxiOut
- TaxiIn
- AirTime
- ActualElapsedTime
- CarrierDelay
- WeatherDelay
- NASDelay
- SecurityDelay
- LateAircraftDelay
- Cancelled
- CancellationCode
- Diverted
- Any variable recorded during or after the flight

These variables may be used in descriptive or diagnostic analysis where appropriate, but not as predictors in the schedule-time model.

For example, delay-cause variables may be used to describe why historical delays occurred, but they cannot be used to predict a delay before it happens.

7. Permitted predictive features

The predictive dataset should primarily include information available before departure.

Calendar features

- Flight date
- Month
- Quarter

- Day of week
- Weekend indicator
- Departure season
- Holiday-period indicator, when derived from an official calendar
- Day of year

Scheduled-time features

- Scheduled departure time
- Scheduled arrival time
- Departure-hour category
- Morning, afternoon, evening, or overnight category
- Scheduled elapsed time

Scheduled times stored in HHMM format must be converted correctly into hour and minute features. They must not be treated as ordinary continuous integers.

Airline and route features

- Reporting airline
- Origin airport
- Destination airport
- Origin–destination route
- Flight distance
- Origin state
- Destination state
- Domestic time-zone difference, where reliably constructed

Historical operational features

Historical aggregate variables may be created, such as:

- Previous delay rate for the airline
- Previous delay rate for the origin airport
- Previous delay rate for the destination airport
- Previous delay rate for the route
- Previous delay rate for an airline–airport combination

- Previous average arrival delay by scheduled departure hour

However, these variables must be calculated using **past data only**.

For example, when predicting flights in 2025, historical airport delay rates may be calculated from 2024 data. The group must not calculate a historical delay rate using records from the test period.

8. Refined research questions

RQ1: Predictive performance

How accurately can schedule-time flight information predict whether a flight will arrive at least 15 minutes late?

RQ2: Delay factors

Which scheduling, airline, airport, route, and temporal factors are most strongly associated with arrival-delay risk?

RQ3: Operational differences

How do delay rates differ across airlines, airports, routes, departure periods, and seasons?

RQ4: Model explainability

How can SHAP explanations help users understand the factors driving overall and individual flight-delay predictions?

RQ5: Operational prioritization

Can an optimization-based prioritization method identify more actually delayed flights than random selection or simple rule-based prioritization when operational capacity is limited?

9. Refined hypotheses

The group should avoid unnecessary hypotheses. Research questions involving dashboards or SHAP do not require formal hypothesis testing.

Suitable hypotheses include:

Airline and airport differences

$$H_0:$$

There is no statistically significant difference in arrival-delay rates across selected airlines or airports.

H_1 :

At least one airline or airport has a significantly different arrival-delay rate.

Predictive model

H_0 :

The predictive model does not perform better than the selected baseline model.

H_1 :

The predictive model performs better than the selected baseline model based on predefined evaluation metrics.

Prioritization

H_0 :

The optimization-based prioritization method does not identify a higher proportion of actually delayed flights than the comparison methods.

H_1 :

The optimization-based prioritization method identifies a higher proportion of actually delayed flights than the comparison methods.

The group must define the baseline and comparison metrics before model testing.

10. Required analytical components

A. Data preparation

The group must complete and document:

- Data acquisition
- Monthly file integration
- Column selection
- Data-type correction
- Duplicate identification
- Missing-value analysis

- Invalid-value detection
- Cancellation and diversion handling
- Outlier review
- Category standardization
- Feature engineering
- Leakage assessment
- Chronological train-validation-test splitting

Cancelled and diverted flights

For the primary arrival-delay prediction model, cancelled and diverted flights should normally be excluded because a standard arrival-delay outcome may not be available or comparable.

They should still be included in separate descriptive analysis.

The report must state clearly:

- How many records were removed
- Why they were removed
- Whether their exclusion could affect interpretation

B. Descriptive analytics

The descriptive component should include a clearly organized set of KPIs.

Required KPIs

- Total flights
- Percentage delayed by at least 15 minutes
- Average arrival delay
- Median arrival delay
- Cancellation rate
- Diversion rate
- Average delay among delayed flights
- Percentage of flights arriving early or on time

Required comparisons

- Delay rate by airline

- Delay rate by origin airport
- Delay rate by destination airport
- Delay rate by route
- Delay rate by month
- Delay rate by day of week
- Delay rate by scheduled departure hour
- Distribution of delay causes
- Cancellation reasons
- Delay trends over time

The team should avoid presenting only raw airline or airport rankings. Results should include flight volume because a small airport with very few flights may appear unusually good or bad.

C. Diagnostic analytics

The purpose of diagnostic analytics is to investigate why delay patterns differ.

Possible methods include:

- Chi-square test of independence
- Cramér's V
- ANOVA
- Kruskal–Wallis test
- Post-hoc comparisons
- Correlation analysis for appropriate numerical variables
- Effect-size calculations
- Logistic-regression coefficient analysis
- Standardized delay-rate comparisons

Examples of diagnostic questions include:

- Is arrival-delay status associated with airline?
- Is arrival-delay status associated with departure-hour category?
- Do average delay minutes differ across airports?
- Are evening flights more likely to be delayed than morning flights?

- Do longer routes have different delay patterns?
- Are observed airline differences partly explained by route and airport mix?

The group must report both statistical significance and practical significance. A very large dataset may produce statistically significant results for effects that are operationally small.

D. Predictive analytics

Required models

The group should develop a limited but meaningful model comparison.

1. **Baseline model**
 - Majority-class or simple historical-rate baseline
2. **Logistic Regression**
 - Interpretable statistical and machine-learning baseline
3. **Random Forest**
4. **One gradient-boosting model**
 - XGBoost or LightGBM

The group does not need to use both XGBoost and LightGBM.

Model-development requirements

- Chronological data split
- Training-only preprocessing
- Class-imbalance assessment
- Hyperparameter tuning
- Cross-validation within the training period
- Validation-set model selection
- One final evaluation on the untouched test set
- Reproducible random seeds where applicable
- Clear comparison with baseline models

Required metrics

Because delayed flights may represent a minority class, accuracy must not be the primary metric.

Report:

- Precision
- Recall
- F1-score
- ROC-AUC
- Precision–Recall AUC
- Confusion matrix
- Brier score or another probability-calibration measure
- Calibration plot
- Lift or recall within the highest-risk group

The selected final model should not be chosen only because it has the highest ROC-AUC. Selection should consider:

- Recall of delayed flights
- Precision
- Probability calibration
- Interpretability
- Operational usefulness
- Computational cost

Error analysis

The group must investigate:

- False positives
- False negatives
- Performance by airline
- Performance by major airport
- Performance by departure period
- Performance by month
- Performance for common and less-common routes

A model with strong overall performance may still be unreliable for particular airports or airlines.

E. Explainable artificial intelligence

SHAP should be applied to the selected final model.

Required global explanations

- SHAP summary plot
- Ranked feature importance
- Direction of feature effects
- Dependence plots for important variables

Required local explanations

The dashboard should show individual explanations for selected flights, such as:

- Scheduled departure time increased the predicted risk
- A historically high-risk route increased the risk
- The airline's previous delay pattern increased or decreased the risk
- The day-of-week pattern affected the result

SHAP must be described as an explanation of the model's prediction, not proof of causation.

The team must not state that a variable "causes" delays solely because it has a high SHAP value.

11. Prescriptive analytics requirement

The original proposal suggests prioritizing high-risk flights using OR-Tools or PuLP but does not fully define the decision problem.

The refined prescriptive problem is:

Given a limited capacity to review or support only K scheduled flights, select the flights with the greatest predicted delay exposure.

Decision variable

$$x_i = \begin{cases} 1, & \text{if flight } i \text{ is selected for operational review} \\ 0, & \text{otherwise} \end{cases}$$

Basic objective

$$\text{Maximize } \sum_{i=1}^n p_i x_i$$

where:

- p_i is the calibrated predicted probability that flight i will be delayed
- x_i indicates whether the flight is selected

Capacity constraint

$$\sum_{i=1}^n x_i \leq K$$

The dashboard should allow the user to change K , such as:

- Review 10 flights
- Review 25 flights
- Review 50 flights
- Review the top 5% of flights

Optional operational constraints

If justified, the group may add constraints such as:

- Maximum number of selected flights per airport
- Minimum number from major operating regions
- Separate capacity limits by time window
- Separate capacity limits by airline

Evaluation of prioritization

The optimization model should be evaluated on the held-out test data.

Compare it with:

- Random selection
- Earliest-departure-first selection
- Selection based on a simple historical delay-rate rule
- Selection based only on the highest predicted probability

Report:

- Percentage of actual delayed flights captured
- Recall within the selected group

- Average actual delay among selected flights
- Lift over random selection
- Number of delayed flights identified per K reviews

The group must not claim that the optimization reduces delays unless actual intervention-effect data are available.

The correct claim is that it improves the **prioritization or identification of high-risk flights**.

12. Dashboard and prototype system

Required dashboard

A Streamlit, Dash, or Power BI dashboard may be used.

Recommended sections are:

Executive overview

- Total flights
- Delay rate
- Average delay
- Cancellation rate
- Major delay patterns

Historical analysis

- Airline comparisons
- Airport comparisons
- Route analysis
- Time-based trends
- Delay-cause analysis

Prediction page

- Flight input fields
- Predicted probability
- Risk category
- Model confidence or calibration information

Explainability page

- Global SHAP results
- Individual SHAP explanation
- Main factors increasing risk
- Main factors decreasing risk

Prioritization page

- User-selected capacity K
- Ranked flight list
- Predicted risk
- Key explanation
- Origin and destination
- Scheduled departure time
- Selection-status indicator

FastAPI

FastAPI should be treated as a **stretch deliverable**, not the central requirement.

A minimal API may contain:

- `/predict`
- `/explain`
- `/prioritize`

The project will not be penalized for using Streamlit directly if the analytics, validation, and dashboard are completed at a high standard.

13. Required final deliverables

The group must submit:

1. Original downloaded BTS data references
2. Data-acquisition documentation
3. Cleaned analytical dataset or reproducible cleaning script
4. Data dictionary
5. Leakage-audit table
6. Exploratory and descriptive analysis

7. Diagnostic statistical analysis
 8. Model-development notebook or scripts
 9. Baseline and advanced model comparison
 10. Final-model evaluation
 11. Calibration analysis
 12. SHAP analysis
 13. Prescriptive prioritization model
 14. Interactive dashboard
 15. Technical documentation
 16. Final written report
 17. Presentation slides
 18. Source-code repository
 19. Individual contribution statement
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14. Minimum success criteria

A successful project must demonstrate that:

- The dataset comes from the official BTS source.
- The study uses a complete and clearly defined period.
- The predictive model uses only information available before departure.
- Data leakage has been explicitly investigated.
- A chronological validation strategy is used.
- At least one baseline and two machine-learning models are compared.
- Model evaluation includes class-imbalance-sensitive metrics.
- Predicted probabilities are evaluated for calibration.
- SHAP is used correctly and not interpreted as causation.
- The prescriptive model is formulated using explicit decision variables, objectives, and constraints.
- The prioritization method is evaluated against at least two baselines.
- The dashboard presents descriptive, predictive, explainable, and prescriptive outputs.

- All results can be reproduced from the submitted code.
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15. Scope boundaries

To keep the project achievable, the following are outside the required scope:

- Real-time airline data streaming
- Live integration with airline operational systems
- Production cloud deployment
- Crew rescheduling
- Aircraft-tail assignment
- Gate-allocation optimization
- Passenger rebooking
- Weather forecasting
- Causal claims about delay reduction
- Full airline network simulation
- Creation of synthetic flight records

Weather data may be added as an optional extension only after the required BTS-based system is complete and only from an official source.

16. Recommended division of responsibilities

With four group members, responsibilities may be distributed as follows:

Role	Main responsibilities
Data and ETL lead	BTS acquisition, integration, cleaning, data dictionary and quality checks
Analytics lead	Descriptive analytics, diagnostic tests, KPIs and visualizations
Machine-learning lead	Feature engineering, modelling, tuning, evaluation and SHAP
Decision-support lead	Optimization, dashboard, system integration and testing

All members must participate in:

- Research design

- Interpretation
- Report writing
- Dashboard review
- Presentation preparation

Each member should understand the complete project rather than only their assigned technical component.

17. Eight-week project timeframe

Week 1 – Business understanding, scope confirmation and data acquisition

Activities

- Finalize the refined title and business problem.
- Define the schedule-time prediction point.
- Confirm the primary target as ArrDel15.
- Select the study years.
- Download the official BTS files.
- Review BTS documentation.
- Select required columns.
- Create the initial data dictionary.
- Estimate data volume and computing requirements.
- Establish the project repository and folder structure.
- Assign group responsibilities.
- Prepare the leakage-audit framework.

Required outputs

- Final project charter
- Final research questions
- Dataset-source documentation
- Selected study period
- Initial data dictionary
- Initial list of permitted and prohibited predictors

- GitHub or equivalent repository
- Team responsibility matrix

Week 1 milestone

The group must be able to explain exactly:

- What is being predicted
 - When the prediction is made
 - What data are available at that time
 - Why partial 2026 data are not being used as the primary study period
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Week 2 – Data integration, cleaning and quality assessment

Activities

- Combine monthly or annual BTS files.
- Standardize column names and data types.
- Convert date and time variables.
- Identify duplicates.
- Analyze missing values.
- Separate completed, cancelled, and diverted flights.
- Inspect invalid delay and time values.
- Assess class distribution.
- Document exclusions.
- Produce the cleaned modelling table.
- Complete the leakage-audit table.
- Establish chronological training, validation, and test periods.

Required outputs

- Reproducible ETL script
- Data-quality report
- Missing-value summary
- Record-exclusion summary

- Final data dictionary
- Leakage-audit table
- Cleaned modelling dataset
- Chronological split specification

Week 2 milestone

The group must have a leakage-safe analytical dataset ready for analysis.

Week 3 – Descriptive analytics and dashboard foundation

Activities

- Calculate required KPIs.
- Analyze delay rate by airline.
- Analyze delay rate by airport.
- Analyze major routes.
- Analyze delays by month, day, and departure hour.
- Examine cancellation and diversion patterns.
- Analyze historical delay causes separately.
- Create clear visualizations.
- Build the initial dashboard structure.
- Identify data-quality or interpretation issues requiring further investigation.

Required outputs

- Descriptive-analysis notebook
- KPI definitions
- At least eight meaningful visualizations
- Airline and airport comparison tables
- Time-pattern analysis
- Preliminary executive dashboard

Week 3 milestone

The group must be able to describe where and when delays are most common without making unsupported causal claims.

Week 4 – Diagnostic analytics and feature engineering

Activities

- Conduct appropriate chi-square tests.
- Calculate Cramér's V or relevant effect sizes.
- Compare delay minutes across selected groups.
- Apply ANOVA or nonparametric alternatives as appropriate.
- Conduct post-hoc analysis where necessary.
- Engineer calendar and scheduled-time features.
- Create route variables.
- Create training-only historical delay-rate features.
- Encode categorical variables.
- Examine multicollinearity.
- Finalize the predictive feature set.
- Build a simple baseline model.

Required outputs

- Diagnostic-analysis report
- Statistical test assumptions
- Effect-size results
- Feature-engineering documentation
- Final feature list
- Baseline model results
- Updated dashboard diagnostic section

Week 4 milestone

The group must have a justified, leakage-free feature set and a baseline against which advanced models can be compared.

Week 5 – Predictive-model development

Activities

- Train Logistic Regression.
- Train Random Forest.
- Train XGBoost or LightGBM.
- Tune selected hyperparameters.
- Address class imbalance where necessary.
- Compare models on validation data.
- Evaluate precision, recall, F1, ROC-AUC, and PR-AUC.
- Generate confusion matrices.
- Assess training time and model complexity.
- Select the candidate final model.
- Preserve the test set without repeated use.

Required outputs

- Model-development pipeline
- Hyperparameter documentation
- Validation-results table
- ROC and precision–recall curves
- Confusion matrices
- Preliminary model-selection justification

Week 5 milestone

The group must select a candidate final model based on operationally meaningful criteria, not accuracy alone.

Week 6 – Final evaluation, calibration, SHAP and error analysis

Activities

- Evaluate the selected model once on the untouched test set.
- Calculate all final metrics.
- Produce a calibration curve.
- Calculate Brier score.
- Apply probability calibration if required.

- Analyze false positives and false negatives.
- Evaluate performance across airlines and airports.
- Conduct SHAP global analysis.
- Create local explanations for selected flights.
- Document model limitations.
- Integrate model results into the dashboard.

Required outputs

- Final test-results table
- Calibration analysis
- Error-analysis report
- Subgroup-performance analysis
- SHAP summary plot
- SHAP dependence plots
- Individual prediction explanations
- Updated prediction and explainability dashboard pages

Week 6 milestone

The group must demonstrate that the model's probabilities are interpretable and that model weaknesses are understood.

Week 7 – Prescriptive prioritization and system integration

Activities

- Formulate the decision variable.
- Define the optimization objective.
- Implement the capacity constraint.
- Add optional justified constraints.
- Test different values of K .
- Compare optimization with random and rule-based methods.
- Calculate lift and delayed-flight capture.
- Create the prioritization dashboard.

- Integrate descriptive, predictive, SHAP, and optimization components.
- Develop minimal FastAPI endpoints only if core components are complete.
- Conduct functional testing.

Required outputs

- Mathematical optimization formulation
- Optimization code
- Baseline-comparison results
- Prioritized flight table
- Scenario analysis for different capacity levels
- Integrated dashboard prototype
- Initial user documentation

Week 7 milestone

A user must be able to select an operational capacity and receive a ranked, explainable list of high-risk flights.

Week 8 – Validation, documentation and presentation

Activities

- Conduct final end-to-end testing.
- Review the project for leakage.
- Verify reproducibility.
- Refine dashboard labels and explanations.
- Complete the final report.
- Prepare management recommendations.
- Prepare technical documentation.
- Finalize presentation slides.
- Rehearse the presentation.
- Prepare responses to methodological questions.
- Complete individual contribution statements.
- Archive code, figures, outputs, and source documentation.

Required outputs

- Final report
- Final dashboard
- Final code repository
- Reproducibility guide
- Technical documentation
- Executive recommendations
- Presentation slides
- Individual contribution statements
- Final demonstration

Week 8 milestone

The completed project must present a connected analytical story:

Historical delay patterns → statistically supported delay factors → leakage-safe predictions → explainable risk scores → optimized operational prioritization.

18. Condensed eight-week schedule

Workstream	W1	W2	W3	W4	W5	W6	W7	W8
Scope and business understanding	•							
Data acquisition and documentation	•	•						
Data cleaning and integration		•						
Descriptive analytics			•	•				
Diagnostic analytics				•				
Feature engineering		•		•	•			
Predictive modelling				•	•	•		
Calibration and error analysis						•		
SHAP explainability						•	•	
Prescriptive optimization							•	

Workstream	W1	W2	W3	W4	W5	W6	W7	W8
Dashboard development			•	•	•	•	•	•
Testing and integration							•	•
Report and presentation	•	•	•	•	•	•	•	•

This refinement retains the original proposal’s comprehensive analytics structure while removing target leakage, narrowing the deployment expectations, and turning the prescriptive component into an evaluable operational-prioritization problem.

Week 1 – Business Understanding and Project Definition

1. Project Charter

Project Title

AI-Powered Flight Delay Risk Prediction and Operational Prioritization Using U.S. Bureau of Transportation Statistics Data

Business Problem

Airline operations managers are responsible for monitoring thousands of scheduled flights every day. Due to limited operational resources, it is not feasible to provide the same level of attention to every flight. Consequently, there is a need for a decision-support system capable of identifying flights with the highest probability of experiencing significant arrival delays before departure.

The proposed solution aims to assist airline operations managers by predicting delay risk, explaining the main contributing factors, and prioritizing flights requiring operational review.

Business Question

Given a set of scheduled flights and limited operational-review capacity, which flights should be prioritized because they have the highest predicted risk of arriving at least 15 minutes late?

Project Purpose

The objective of this project is to develop an AI-powered decision support system capable of:

- Analyzing historical airline delay patterns.
- Identifying operational factors associated with arrival delays.
- Predicting whether a scheduled flight will arrive at least 15 minutes late.
- Explaining model predictions using SHAP.
- Prioritizing high-risk flights when operational capacity is limited.
- Presenting results through an interactive dashboard.

2. Project Scope

In Scope

- Historical analysis of BTS airline operations data.
- Flight delay prediction.
- Explainable AI using SHAP.
- Flight prioritization through optimization.
- Interactive dashboard.

Out of Scope

- Real-time airline integration.
- Weather forecasting.
- Crew scheduling.
- Passenger rebooking.
- Aircraft assignment.
- Cloud deployment.

3. Dataset Documentation

Dataset Source

Dataset Name

Airline On-Time Performance Data

Publisher

U.S. Bureau of Transportation Statistics (BTS)

Dataset URL

<https://www.transtats.bts.gov/ONTIME/>

Study Period

(To be completed after the team agrees.)

Example

Training

2024

Testing

2025

4. Research Questions

RQ1

How accurately can schedule-time flight information predict whether a flight will arrive at least 15 minutes late?

RQ2

Which scheduling, airline, airport, route, and temporal factors are most strongly associated with arrival-delay risk?

RQ3

How do delay rates differ across airlines, airports, routes, departure periods, and seasons?

RQ4

How can SHAP explanations improve the interpretation of flight-delay predictions?

RQ5

Can optimization techniques prioritize high-risk flights more effectively than simple selection methods?

5. Target Variable

Primary Target

ArrDel15

Binary Classification

- 1 = Arrival delay $\geq$ 15 minutes
- 0 = Arrival delay < 15 minutes

Prediction is made **before departure**, using only information available during flight scheduling.

6. Permitted Predictive Variables

Category	Variables
Calendar	FlightDate, Month, Quarter, DayOfWeek
Schedule	CRSDepTime, CRSArrTime, CRSElapsedTime
Airline	Reporting_Airline
Route	Origin, Destination, Distance
Derived Features	Departure Hour, Season, Weekend, Historical Delay Rates

7. Prohibited Predictors (Leakage Audit)

Variable	Reason
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DepDelay	Known after departure
TaxiOut	Known after departure
TaxiIn	Known after landing
ArrDelay	Target outcome
CarrierDelay	Recorded after delay occurs
WeatherDelay	Recorded after delay occurs
Cancelled	Unknown before departure
Diverted	Unknown before departure

8. Team Responsibilities

Member	Responsibility
Daniel	Machine Learning, SHAP, Report
Christian	Data Preparation & ETL
Alexis	Dashboard & Visualization
Navpreet	Optimization & Documentation

(Update as needed.)

Week 2 – Data Preparation

This section should remain mostly empty until you've completed the work.

9. Data Acquisition

- Download BTS monthly files.
- Document downloaded datasets.
- Record download dates.

10. ETL Process

(To be completed.)

- Merge monthly datasets.
- Convert data types.
- Handle missing values.
- Standardize column names.

11. Data Quality Assessment

Document:

- Duplicate records
- Missing values
- Invalid values
- Cancelled flights
- Diverted flights

12. Record Exclusions

Example table

Reason	Records Removed
Cancelled Flights	
Diverted Flights	
Missing Target	

13. Final Data Dictionary

This will become one appendix.

14. Chronological Split

Example

Training

January–December 2024

Validation

January–September 2025

Testing

October–December 2025

15. Week 2 Deliverables Checklist

- Dataset downloaded
- ETL completed
- Data quality report
- Leakage audit
- Final data dictionary
- Chronological split
- Clean analytical dataset

I think we should treat this as a real consulting engagement

Rather than simply writing sections to satisfy the course requirements, we can build the project incrementally. At the end of each week, we'll complete the corresponding deliverables and integrate them into the final report. That way, by Week 8, the report will be almost finished instead of being written all at once. This approach aligns well with your professor's timeline and makes the project much easier to manage.